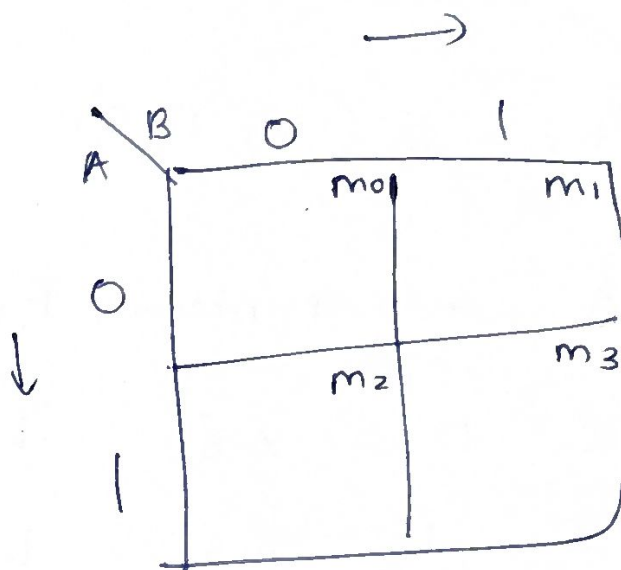


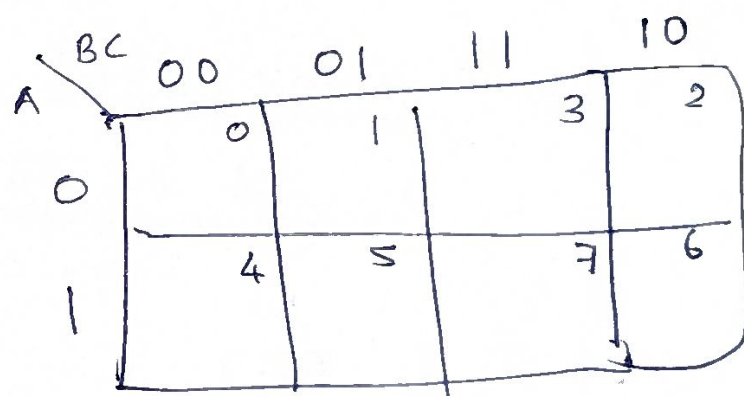
2 variable K-Map

A	B	Min term	
0	0	$\bar{A}\bar{B}$	m_0
0	1	$\bar{A}B$	m_1
1	0	$A\bar{B}$	m_2
1	1	AB	m_3



3 variable K Map

A	B	C	Min terms	
0	0	0	$\bar{A}\bar{B}\bar{C}$	m_0
0	0	1	$\bar{A}\bar{B}C$	m_1
0	1	0	$\bar{A}B\bar{C}$	m_2
0	1	1	$\bar{A}BC$	m_3
1	0	0	$A\bar{B}\bar{C}$	m_4
1	0	1	$A\bar{B}C$	m_5
1	1	0	$AB\bar{C}$	m_6
1	1	1	ABC	m_7



Q2

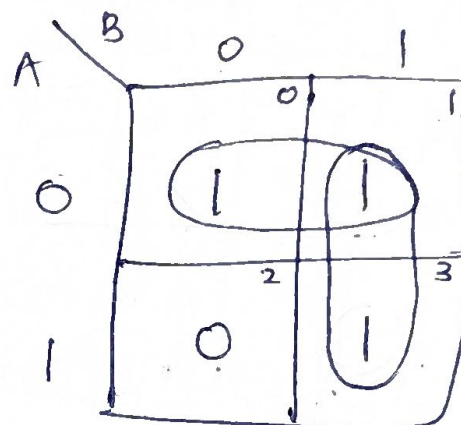
Group of
power of 2
2, 4, 8, 16

2 variable K-Map

Gg①

$$F_1 = \sum (0, 1, 3)$$

A	B	Minterms	F_1
0	0	$\bar{A}\bar{B}$	1
0	1	$\bar{A}B$	1
1	0	$A\bar{B}$	0
1	1	AB	1



$$F_1 = \bar{A}\bar{B} + \bar{A}B + AB$$



Variable fixed

Discard the variable changing

$$\bar{A}\bar{B} + \bar{A}B$$

$$= \bar{A}(\bar{B} + B)$$

$$= \bar{A}$$



$$\bar{A}B + AB$$

$$B(\bar{A} + A)$$

$$= B$$

$$\underline{\underline{F_1 = \bar{A} + B}}$$

Eg ②

$$f_1 = \sum (0, 3)$$

~~f_1 = \sum (0, 1, 2)~~

A B

A \ B	0	1
0	1	0
1	0	1

Eg ③

$$f_1 = \underline{\underline{\bar{A}\bar{B} + AB}}$$

$$f_1 = \sum (1, 2)$$

$$f_1 = \bar{A}B + A\bar{B}$$

A \ B	0	1
0		1
1	1	

Eg ④

A \ B	0	1
0	1	1
1	1	1

$$\underline{\underline{F = 1}}$$

3 variable K-Map

Eg ①

$$f_1 = \sum (0, 1, 5)$$

A \ BC	00	01	11	10
0	1	1		
1		1		

$$f_1 = \bar{A}\bar{B} + \bar{B}C$$

~~Eg ①~~

Eg ② $f_1 = \Sigma(0, 1, 5, 7)$

	BC	00	01	11	10
A					
0		0	1	3	2
1		4	5	7	6

$f_1 = \bar{A}\bar{B} + AC$

Eg ③ $f_1 = \Sigma(0, 1, 3, 5)$

	BC	00	01	11	10
A					
0		0	1	3	2
1		4	5	7	6

$f_1 = \bar{A}\bar{B} + \bar{B}C + \bar{A}C$

Eg ④ $f_1 = \Sigma(0, 2)$

	BC	00	01	11	10
A					
0		0	1	3	2
1		4	5	7	6

$f_1 = \bar{A}\bar{C}$

$f_1 = \Sigma(4, 6) = A\bar{C}$

Eg ⑤

$$F_1 = \sum (1, 3, 5, 7)$$

		BC			
		00	01	11	10
A	0	0	1	3	2
	1	4	5	7	6

$$F_1 = C$$

Eg ⑥

$$F_1 = \sum (0, 2, 4, 6)$$

		BC			
		00	01	11	10
A	0	1		3	2
	1	4	5	7	6

$$F_1 = \bar{C}$$

Eg ⑦

$$F_1 = \sum (0, 1, 2, 3)$$

		BC			
		00	01	11	10
A	0	1	1	3	2
	1	4	5	7	6

$$F_1 = \bar{A}$$

Eg ⑧

$$F_1 = \sum (4, 5, 6, 7)$$

$$F_1 = A$$

⑨ $f_1 = \sum (0, 1, 3, 4, 5, 7)$

A	BC		00	01	11	10
			0	1	3	2
0			1	1	1	
1			1	1	1	

$f_1 = \bar{B} + C$

⑩ $f_1 = \sum (0, 1, 2, 4, 5, 6)$

A	BC		00	01	11	10
			0	1	3	2
0			1	1		1
1			1	1		1

$f_1 = \bar{B} + \bar{C}$

4 Variable K-Map

5

$$f_1(A, B, C, D)$$

$$2^4 = 16$$

Minterms

AB \ CD	00	01	11	10
00	0 1	1 1	3 1	2
01	4 1	5	7	6
11	12	13 1	15 1	14
10	8 1	9	11	10

$$f_1 = \sum (0, 1, 3, 4, 8, 13, 15)$$

$$f_1 = \begin{matrix} A\bar{B}\bar{C}D & + & \bar{A}B\bar{C}D & + & \bar{A}\bar{B}CD & + & \bar{A}\bar{B}\bar{C}\bar{D} & + \\ 1001 & & 0101 & & 0010 & & 0000 & m_0 \\ m_9 & & m_5 & & m_2 & & ABCD & \\ (Algebraic form) & & & & & & 1111 & m_{15} \end{matrix}$$

Rules

- 1) Make the group of adjacent 1s (Group of 2, 4, 8 etc)
- 2) Try to cover all the 1s using minimum number of groups.
- 3) Overlapping of the group is allowed during minimization.
- 4) Consider those minterms individually, which cannot be grouped with other minterms.

$$F_1(A, B, C, D)$$

A	B	C	D	
0	0	0	0	m_0
0	0	0	1	m_1
0	0	1	0	m_2
0	0	1	1	m_3
0	1	0	0	m_4
0	1	0	1	m_5
0	1	1	0	m_6
0	1	1	1	m_7
1	0	0	0	m_8
1	0	0	1	m_9
1	0	1	0	m_{10}
1	0	1	1	m_{11}
1	1	0	0	m_{12}
1	1	0	1	m_{13}
1	1	1	0	m_{14}
1	1	1	1	m_{15}

Eg ① $\Sigma (0, 2, 5, 9, 15)$

AB \ CD	00	01	11	10
00	0 1	1	3	2 1
01	4	5 1	7	6
11	12	13	15 1	14
10	8	9 1	11	10

$$\bar{A}\bar{B}\bar{D} + \bar{A}B\bar{C}D +$$

$$ABCD + A\bar{B}\bar{C}D$$

Eg ②

$$\Sigma (4, 5, 6, 9, 13, 14)$$

AB \ CD	00	01	11	10
00				
01	1	1		1
11		1		1
10		1		

$$\bar{A}B\bar{C} + A\bar{C}D + BCD$$

Eg ③

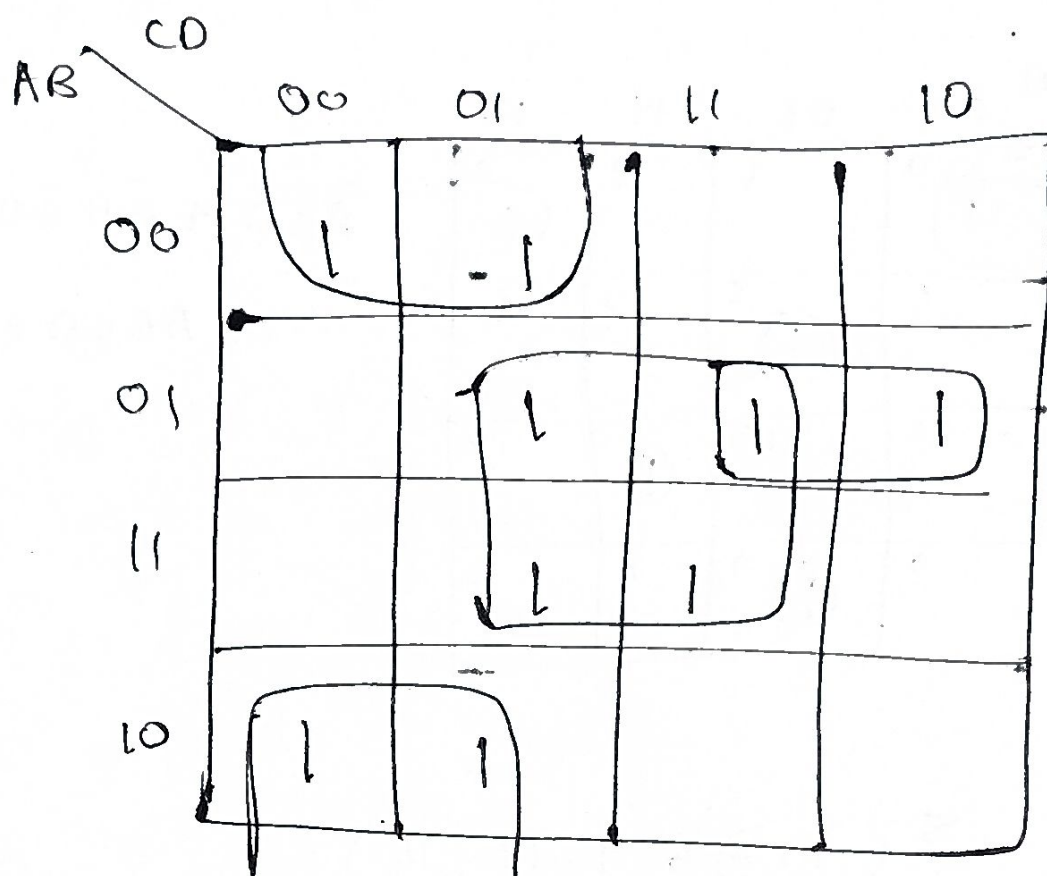
$$\Sigma (4, 5, 6, 7, 9, 13, 14, 15)$$

AB \ CD	00	01	11	10
00				
01	1	1	1	1
11		1	1	1
10		1		

$$\bar{A}B + BC + A\bar{C}D$$

Eg (4)

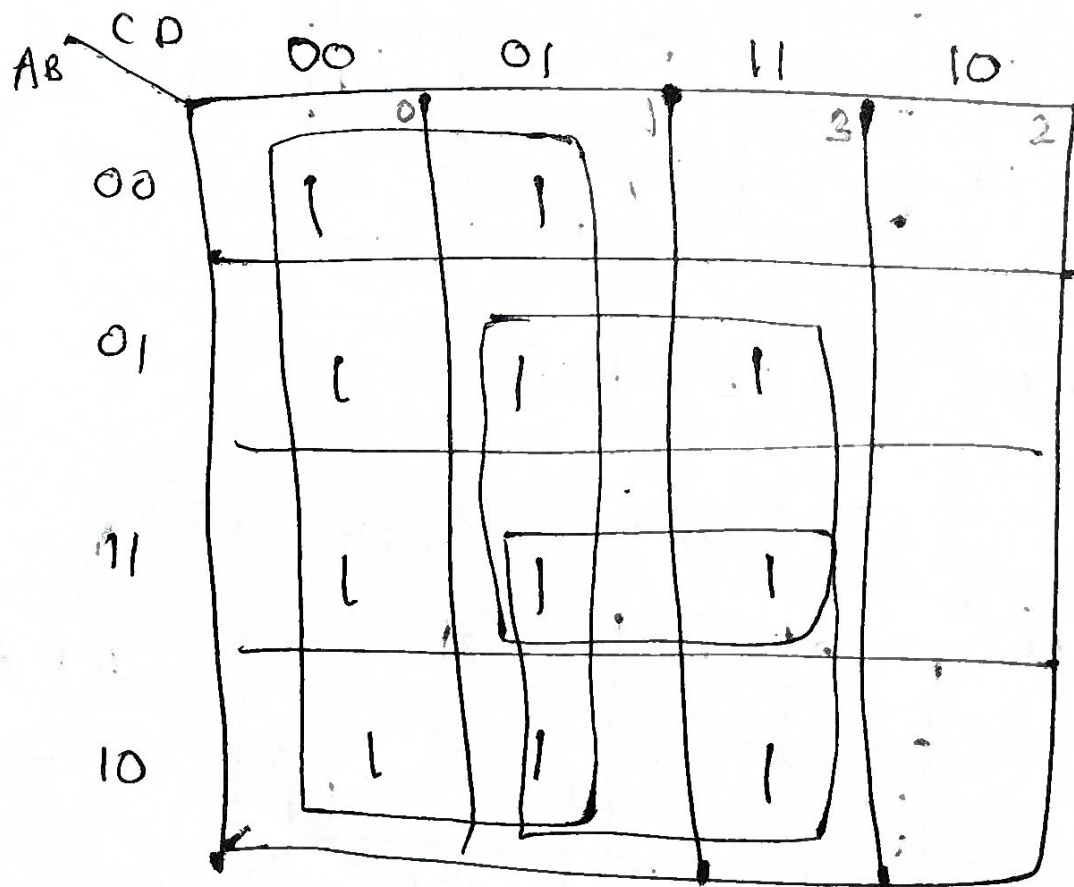
$$\Sigma(0, 1, 5, 6, 7, 8, 9, 13, 15)$$



$$F = \bar{B}\bar{C} + BD + \bar{A}BC$$

Eg (5)

$$F = \Sigma(0, 1, 4, 5, 7, 8, 9, 11, 12, 13, 15)$$



$$F = \bar{C} + AD + BD$$